

Phase transitions in sparse PCA: the near optimality of soft thresholding

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joint work with Theodor Misiakiewicz and Elad Romanov

Low-rank and sparse matrices

- Principal Component Analysis (PCA) is a standard tool for low-rank signal recovery.
 - Key drawback: each principal component (singular vector) is a linear combination of all original variables.
- Often, signal matrices are simultaneously low-rank and sparse.

Low-rank and sparse matrices

- Principal Component Analysis (PCA) is a standard tool for low-rank signal recovery.
 - Key drawback: each principal component (singular vector) is a linear combination of all original variables.
- Often, signal matrices are simultaneously low-rank and sparse.
- J. Friedman et al. (2004) advises “betting on sparsity”:

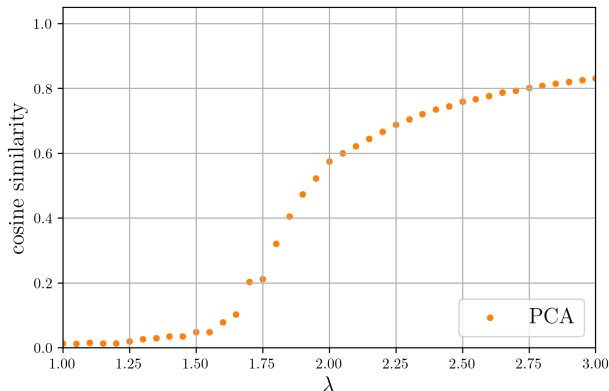
Use a procedure that does well in sparse problems, since no procedure does well in dense problems.

PCA is inconsistent in high dimensions

- Observe $\mathbf{x}_1, \dots, \mathbf{x}_n \stackrel{i.i.d.}{\sim} \mathcal{N}(0, \mathbf{\Sigma})$, $\mathbf{\Sigma} := (\lambda - 1)\mathbf{v}\mathbf{v}^\top + \mathbf{I}_p$.

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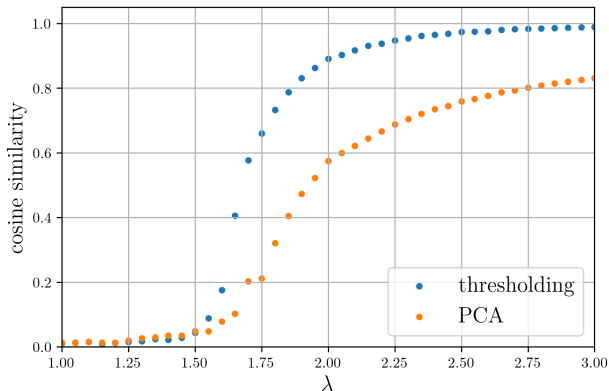
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- Here, $n = 10,000$ and $p = 5,000$.

PCA with thresholding preprocessing

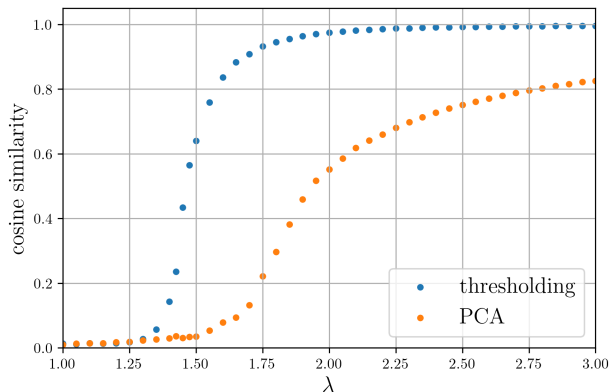
- Soft-threshold $\mathbf{S} := n^{-1} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^\top$ elementwise.



- Here, $n = 10,000$, $p = 5,000$, and $m := \|\mathbf{v}\|_0 = 50$.

PCA with thresholding preprocessing

- Performance of thresholding improves with increasingly sparse $\mathbf{v}$.



- Here, $n = 10,000$, $p = 5,000$, and $m := \|\mathbf{v}\|_0 = 25$.

Objectives for this talk

- Investigate impact of preprocessing on estimation of $\boldsymbol{v}$.
- Understand phase transition phenomena:

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- Investigate impact of preprocessing on estimation of $\mathbf{v}$.
- Understand phase transition phenomena:
 - Spiked model is fundamental—theory must explain the prior plots.
 - Phase transitions justify the scree plot “elbow” method.
 - Determine the relative signal strengths, sparsity levels, and sample sizes required for estimation.

Sparse PCA methods

- SCoTLASS—Jolliffe and Uddin (2003).
- diagonal thresholding—Johnstone and Lu (2004).
- elastic net regression—Zou, Hastie, Tibshirani (2006).
- semidefinite programming—d’Aspremont et al. (2007)
- fantope projection and selection—Vu et al. (2013)
- covariance thresholding—Krauthgamer et al. (2015).
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Sparse PCA model

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- Form the sample covariance matrix

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- Let $m := \|\mathbf{v}\|_0$ denote the sparsity level and

$$v_1, \dots, v_m = \pm \frac{1}{\sqrt{m}}, \quad v_{m+1} = \dots = v_p = 0.$$

- Our high-dimensional framework: $n, p, m \rightarrow \infty$,

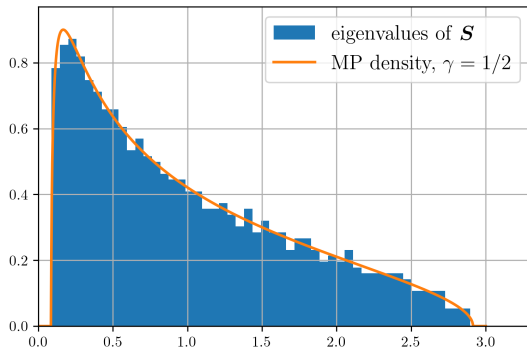
$$p/n \rightarrow \gamma > 0, \quad \frac{m}{\sqrt{n}} \rightarrow \beta > 0.$$

PCA in high-dimensions

- As $p/n \rightarrow \gamma \in [0, 1)$, eigenvalues of $\mathbf{S}$ converges to the Marchenko–Pastur law, with support $[(1 - \sqrt{\gamma})^2, (1 + \sqrt{\gamma})^2]$.

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- Here, $n = 2,000$, $p = 1,000$, so $\gamma = 1/2$.

Eigenvector inconsistency

- Let $\lambda_1, \mathbf{u}_1$ denote the first eigenpair of $\mathbf{S}$.
- λ_1 is biased upwards from λ (Baik and Silverstein, 2006):

$$\lambda_1 \xrightarrow{a.s.} \begin{cases} \lambda + \frac{\lambda\gamma}{\lambda - 1} & \lambda > 1 + \sqrt{\gamma} \\ (1 + \sqrt{\gamma})^2 & \lambda \leq 1 + \sqrt{\gamma} \end{cases}.$$

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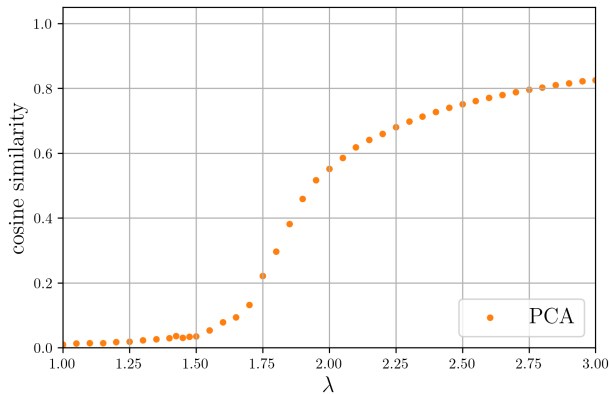
- $\mathbf{u}_1$ is an inconsistent estimator of $\mathbf{v}$ (Paul, 2007):

$$\langle \mathbf{u}_1, \mathbf{v} \rangle^2 \xrightarrow{a.s.} \begin{cases} \frac{1 - \gamma/(\lambda-1)^2}{1 + \gamma/(\lambda-1)} & \lambda > 1 + \sqrt{\gamma} \\ 0 & \lambda \leq 1 + \sqrt{\gamma} \end{cases}.$$

- PCA exhibits a phase transition at $1 + \sqrt{\gamma}$.

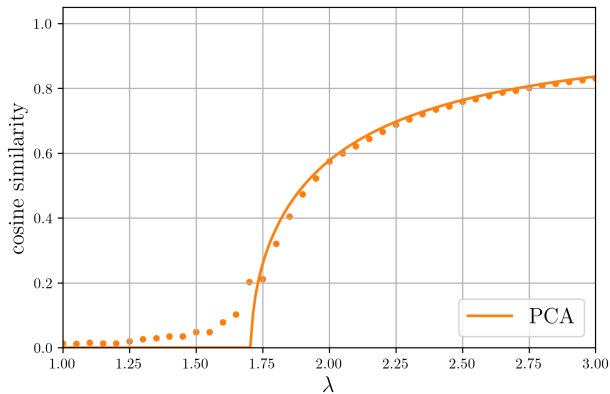
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The “critical” sparsity regime

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$$v_1, \dots, v_m = \pm \frac{1}{\sqrt{m}}, \quad v_{m+1} = \dots = v_p = 0.$$

- $\sqrt{n}\mathbf{S}_{ij} \xrightarrow{d} \mathcal{N}(0, 1)$ for $i, j > m$ and $i \neq j$.

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$$\mathbf{S} - \mathbf{I}_p = \underbrace{\begin{bmatrix} \tilde{O}_p\left(\frac{1}{m}\right) & \tilde{O}_p\left(\frac{1}{\sqrt{mn}}\right) \\ \tilde{O}_p\left(\frac{1}{\sqrt{mn}}\right) & 0 \end{bmatrix}}_{m \quad p-m} + \begin{bmatrix} \tilde{O}_p\left(\frac{1}{\sqrt{n}}\right) & \tilde{O}_p\left(\frac{1}{\sqrt{n}}\right) \\ \tilde{O}_p\left(\frac{1}{\sqrt{n}}\right) & \tilde{O}_p\left(\frac{1}{\sqrt{n}}\right) \end{bmatrix}$$

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- $m \ll \sqrt{n}$: diagonal thresholding.
- $m \gg \sqrt{n}$: no poly-time algorithm recovers $\text{supp}(\mathbf{v})$.
- $m \asymp \sqrt{n}$: signal and noise are comparable.

Covariance thresholding (CT)

- Krauthgamer et al. (2015) proposed covariance thresholding:
 - Soft-threshold $\sqrt{n}\mathbf{S}$ elementwise at level t_1 ,
 - Compute the maximum eigenvector $\mathbf{u}_1$,
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Theorem 1 (Deshpande, Montanari (2015))

There exist $t_1, t_2 > 0$ such that with probability $1 - o(1)$,

$$|\langle \mathbf{u}_1, \mathbf{v} \rangle|^2 \geq 1 - C_1 \cdot \frac{m}{\sqrt{n}} \cdot \frac{\lambda^2 \vee 1}{\lambda^2},$$

and CT recovers $\mathbf{v}$ if $m \leq C_2(\lambda)\sqrt{n}$.

Generalized covariance thresholding (GCT)

- Given $f : \mathbb{R} \mapsto \mathbb{R}$, estimate $\mathbf{v}$ by the first eigenvector $\mathbf{u}_1$ of

$$(\mathbf{K}(f))_{ij} := \begin{cases} n^{-1/2} f(\sqrt{n} \mathbf{S}_{ij}) & i \neq j \\ 0 & i = j \end{cases}.$$

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- Theorems involve the related matrix

$$(\mathbf{K}_0(f))_{ij} := \begin{cases} n^{-1/2} f(\sqrt{n} (\boldsymbol{\Sigma}^{-1/2} \mathbf{S} \boldsymbol{\Sigma}^{-1/2})_{ij}) & i \neq j \\ 0 & i = j \end{cases}.$$

- Hermite polynomials $\{h_k\}_{k \in \mathbb{N}_0}$, coefficients $\{a_k\}_{k \in \mathbb{N}_0}$:

$$a_k := \frac{1}{\sqrt{2\pi}} \int f(x) h_k(x) e^{-x^2/2} dx.$$

Approximating $\mathbf{K}(f)$

Theorem 2

Let f be odd, continuous, a.e. differentiable, and $f''(x)$ exponentially bounded. As $n \rightarrow \infty$ with $p/n \rightarrow \gamma$ and $m/\sqrt{n} \rightarrow \beta$,

$$\|\mathbf{K}(f) - \mathbf{V}\mathbf{\Lambda}\mathbf{V}^\top + \mathbf{K}_0(f)\|_2 \xrightarrow{a.s.} 0,$$

where $\mathbf{V} := [\mathbf{v}, \mathbf{\Sigma}^{-1/2}\mathbf{S}\mathbf{\Sigma}^{-1/2}\mathbf{v}]$,

$$\mathbf{\Lambda} := \begin{bmatrix} a_1(\sqrt{\lambda} - 1)^2 + \tau & a_1(\sqrt{\lambda} - 1) \\ a_1(\sqrt{\lambda} - 1) & 0 \end{bmatrix}, \quad \tau := \sum_{k=2}^{\infty} \frac{a_k \lambda^k}{\beta^{k-1} \sqrt{k!}}.$$

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- Recall $\mathbf{K}_0(f)$ is the kernel matrix of $\mathbf{\Sigma}^{-1/2}\mathbf{S}\mathbf{\Sigma}^{-1/2}$.
- Spectral distribution of $\mathbf{K}_0(f)$ converges to the free convolution of the MP and Wigner semicircle laws (Chen and Singer, 2013).

Spectral properties of $K(f)$

Theorem 3 (informal)

There exists $\rho(f, \gamma, \beta) > 1$ and function $\theta(\lambda; f, \gamma, \beta)$ such that

$$\langle \mathbf{u}_1, \mathbf{v} \rangle^2 \xrightarrow{a.s.} \begin{cases} \theta^2(\lambda; f, \gamma, \beta) & \lambda > \rho(f, \gamma, \beta) \\ 0 & \lambda \leq \rho(f, \gamma, \beta) \end{cases}.$$

Moreover, if $\lambda > \rho(f, \gamma, \beta)$, for any $\varepsilon > 0$,

$$u_{1i} = \theta(\lambda; f, \gamma, \beta)v_i + O_{a.s.}(n^{-1/2+\varepsilon}), \quad 1 \leq i \leq p.$$

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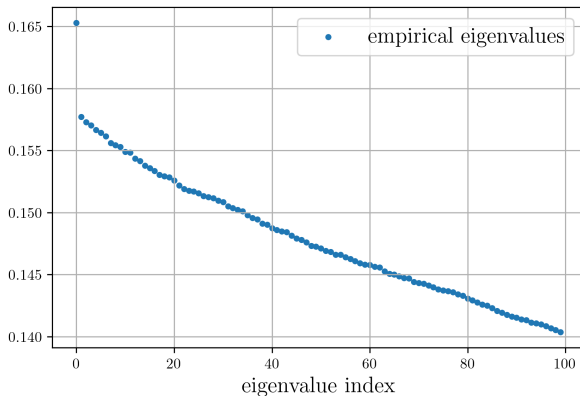
$$u_{1i} = \theta(\lambda; f, \gamma, \beta)v_i + O_{a.s.}(n^{-1/2+\varepsilon}), \quad 1 \leq i \leq p.$$

Corollary 4

Let $\alpha \in (1/2, 1/4)$. If $\lambda > \rho(f, \gamma, \beta)$, hard-thresholding $\mathbf{u}_1$ at level $n^{-\alpha}$ and rounding recovers $\mathbf{v}$ almost surely.

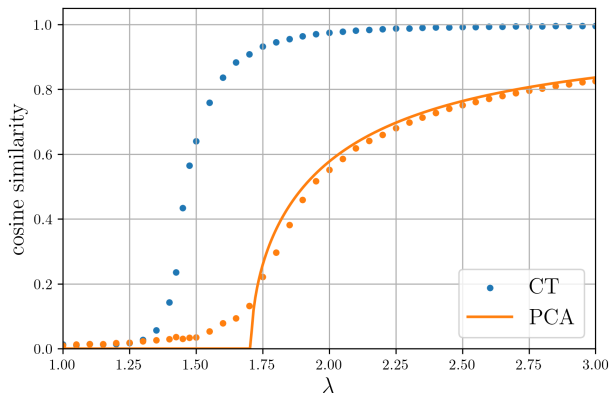
Covariance thresholding

- CT with $n = 10,000$, $p = 5,000$, $m = 25$, $\lambda = 1.48$, and $t_1 = 2$.



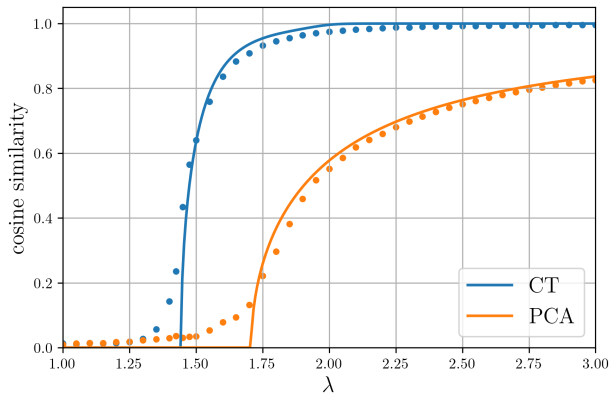
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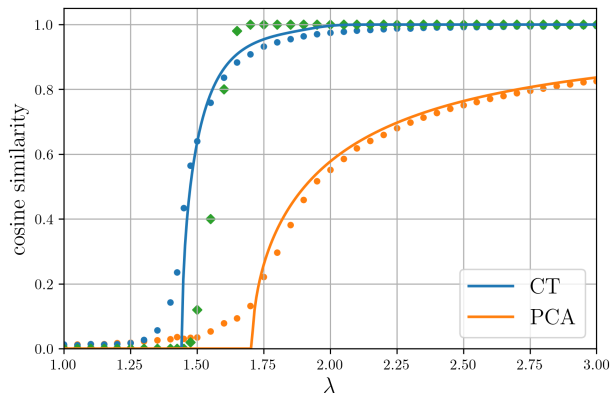
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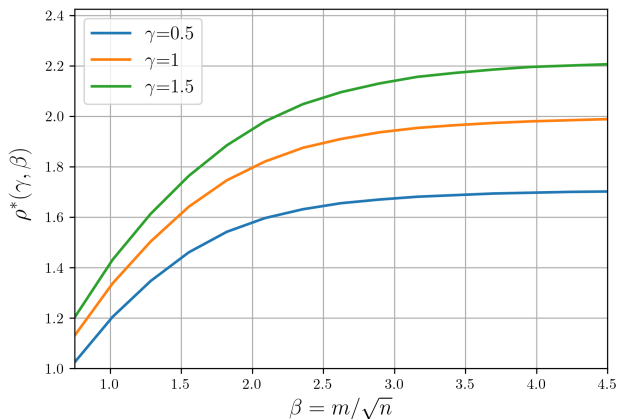
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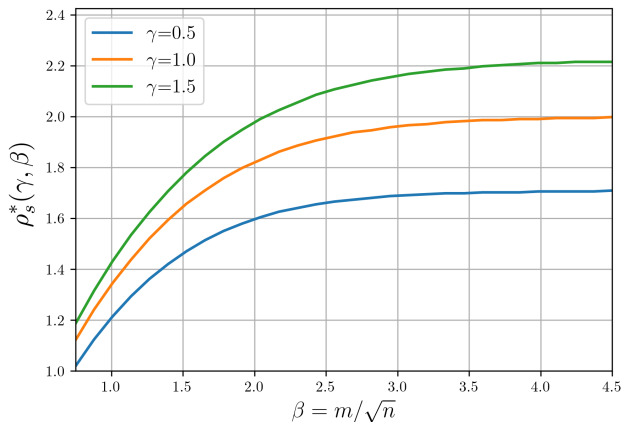
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Soft thresholding is nearly optimal

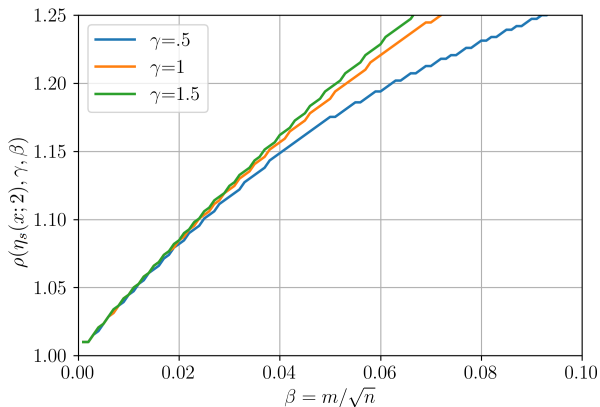
- Let $\eta_s(x; t) := \text{sign}(x)(|x| - t)$; we numerically calculate

$$\rho_s^*(\gamma, \beta) = \inf_{t \geq 0} \rho(\eta_s(x; t), \gamma, \beta).$$



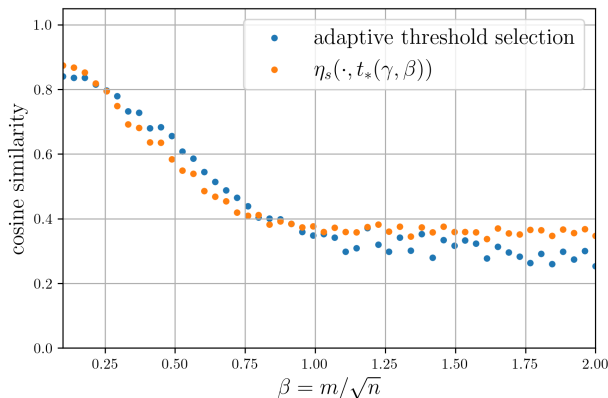
Soft thresholding, $\beta \rightarrow 0$

- As $\beta \rightarrow 0$, $\rho(\eta_s(x; 2), \gamma, \beta)$ decreases linearly:



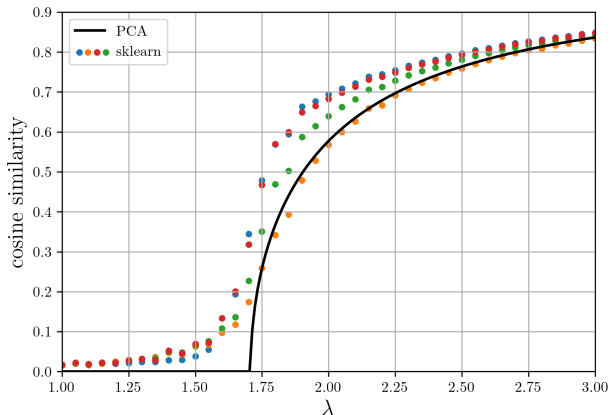
Adaptive threshold selection

- The optimal thresholding level $t_*(\gamma, \beta)$ depends on β .
- Adaptively select t to maximize $(\lambda_1 - \lambda_2)/\lambda_2$:



Zou, Hastie, Tibshirani (2006)

- `sklearn.decomposition.SparsePCA( $\alpha$ )`, $\alpha \in \{1, 2, 3, 4\}$:



Theorem 2: proof sketch

- Let $\mathbf{X} := \mathbf{S} - \mathbf{I}_p - \boldsymbol{\Sigma}^{-1/2} \mathbf{S} \boldsymbol{\Sigma}^{-1/2}$.
- Consider a polynomial kernel f of degree L :

$$\mathbf{K}(f) = \sum_{\ell=0}^L \frac{1}{\ell!} (\sqrt{n} \mathbf{X})^{\odot \ell} \odot \mathbf{K}_0(f^{(\ell)}),$$

where $\odot$ denotes the Hadamard (elementwise) product.

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where $\odot$ denotes the Hadamard (elementwise) product.

- If $\mathbf{v}$ is dense ($m \gg \sqrt{n}$), $\|(\sqrt{n} \mathbf{X})^{\odot \ell}\|_2 \xrightarrow{a.s.} 0$ for $\ell > 1$ and

$$\mathbf{K}(f) \approx \mathbf{K}_0(f) + \sqrt{n} \mathbf{X} \odot \mathbf{K}_0(f').$$

Theorem 2: proof sketch

- Expand $\mathbf{K}(f)$ in terms of Hermite polynomials:

$$\mathbf{K}(f) = \sum_{\ell=1}^L \sum_{k=1}^{L-\ell} \frac{a_{\ell+k}}{k!} \sqrt{\frac{(\ell+k)!}{\ell!}} \left[(\sqrt{n}\mathbf{X})^{\odot k} \odot \mathbf{K}_0(h_\ell) \right],$$

$$(\mathbf{K}_0(h_\ell))_{ij} := \begin{cases} n^{-1/2} h_\ell(\sqrt{n}(\boldsymbol{\Sigma}^{-1/2} \mathbf{S} \boldsymbol{\Sigma}^{-1/2})_{ij}) & i \neq j \\ 0 & i = j \end{cases}.$$

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- Since $\sqrt{n}(\boldsymbol{\Sigma}^{-1/2} \mathbf{S} \boldsymbol{\Sigma}^{-1/2})_{ij} \xrightarrow{d} \mathcal{N}(0, 1)$, $\{h_\ell\}_{\ell \in \mathbb{N}_0}$ orthonormal:
 - $\mathbf{K}_0(h_0), \mathbf{K}_0(h_1), \dots, \mathbf{K}_0(h_L)$ are approximately independent.
 - $\|(\sqrt{n}\mathbf{X})^{\odot k} \odot (\mathbf{K}_0(h_\ell))\|_2 \lesssim n^{-1/4}$ if $\ell, k \geq 1$.

Theorem 3: the “master equation” approach

- $\mathbf{K}(f) \approx \mathbf{V}\mathbf{\Lambda}\mathbf{V}^\top + \mathbf{K}_0(f)$; define

$$\mathbf{R}(z) := (\mathbf{K}_0(f) - z\mathbf{I}_p)^{-1}.$$

- If λ_1 is an outlier, it solves

$$\det(\mathbf{\Lambda}^{-1} - \mathbf{V}^\top \mathbf{R}(\lambda_1) \mathbf{V}) = 0.$$

- Let $\mathcal{C} \subset \mathbb{C}$ be a contour encircling only λ_1 :

$$u_{1i}^2 = \frac{1}{2\pi i} \oint_{\mathcal{C}} \mathbf{e}_i^\top \mathbf{R}(z) \mathbf{V}^\top (\mathbf{\Lambda}^{-1} + \mathbf{V}^\top \mathbf{R}(z) \mathbf{V})^{-1} \mathbf{V} \mathbf{R}(z) \mathbf{e}_i dz.$$

- Similar formulas in Kato's perturbation theory text (1966).

Conclusion

- In high dimensions, generalized covariance thresholding experiences a phase transition at $\rho(f, \gamma, \beta)$.

